23A01604b • OPEN CHANNEL FLOW

UNIT – 2

Steady Varied Flows in Open Channels

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Basic Concepts of Uniform Flow
- Specific Energy and Specific Force Concepts
- Dynamic Equation for Spatially Varied Flows
- Flow Profile Computations
- Introduction to HEC-RAS
- Spatially Varied Flows and Rapidly Varied Flows – Applications

1. BASIC CONCEPTS OF OPEN CHANNEL FLOW

1.1 Open channel vs pipe flow

An **open channel** is a conduit in which a liquid flows with a **free surface** exposed to atmospheric pressure. The driving force is gravity (component of weight along the bed slope), whereas in pipes it is the pressure gradient.

Feature	Pipe flow	Open-channel flow
Driving force	Pressure gradient	Gravity (bed slope)
Free surface	Absent (flows full)	Present; p = 0 gauge
Flow section	Fixed geometry	Depth varies with Q, so section varies
Roughness	Fairly constant	Varies with depth, bed forms, vegetation
Analysis	Easier	Harder: unknown depth is part of the solution

1.2 Channel geometry

- **Prismatic channel:** constant cross-section and bed slope. **Non-prismatic:** natural rivers.
- **Depth y:** vertical distance from the lowest point of the section to the free surface. **Flow depth normal to bed** d = y cos θ.
- **Top width T:** width at free surface. **Area A. Wetted perimeter P. Hydraulic radius R = A/P. Hydraulic depth D = A/T.**
- **Section factor** for critical flow $Z = A\sqrt{D} = A\sqrt{(A/T)}$; for uniform flow $AR^{2/3}$.

Section	Area A	Wetted perimeter P	Top width T
Rectangle (b, y)	by	b + 2y	b
Trapezoid (b, y, side slope m:1)	(b + my)y	$b + 2y\sqrt{(1 + m^2)}$	b + 2my
Triangle (m, y)	my^2	$2y\sqrt{(1 + m^2)}$	2my
Circle (D, central angle θ)	$(D^2/8)(\theta - \sin\theta)$	Dθ/2	D sin(θ/2)
Parabola	(2/3)Ty	$T + 8y^2/3T$	—

1.3 Classification of open-channel flow

- **Steady vs unsteady** (time).
- **Uniform vs varied** (space). Varied flow is subdivided into **gradually varied flow (GVF)** where depth changes slowly over long distances (e.g. backwater curves), and **rapidly varied flow (RVF)** where depth changes abruptly over short distance (hydraulic jump, flow over a weir, sluice-gate flow).
- **Spatially varied flow (SVF):** discharge changes along the channel because of lateral inflow or outflow (side-weir, gutter, wash-water trough).
- **Subcritical, critical and supercritical:** depends on Froude number

$$\mathbf{Fr = V / \sqrt{(g\,D)} = V / \sqrt{(g\,A/T)}}$$

Fr < 1 subcritical (tranquil), Fr = 1 critical, Fr > 1 supercritical (rapid). Information (small surface waves of speed $c = \sqrt{(gD)}$) can travel upstream in subcritical flow only; hence **subcritical flow is controlled from downstream and supercritical flow from upstream**.

- **Laminar vs turbulent:** Re = VR/ν; laminar < 500, turbulent > 2000 (practically always turbulent).

1.4 Pressure distribution and velocity distribution

Pressure is **hydrostatic** for channels of small slope and parallel streamlines: $p = \gamma(y - z)$. The velocity varies across the section; maximum velocity occurs slightly below the free surface (about 0.05–0.25 y). The mean velocity is approximately the velocity at 0.6 y from the surface, or the average of velocities at 0.2 y and 0.8 y. **Energy correction factor α** = $(1/AV^3)\int v^3 dA \approx 1.03–1.36$; **momentum correction β** ≈ 1.01–1.12. Usually taken as 1.

2. UNIFORM FLOW

2.1 Definition and conditions

Uniform flow occurs when depth, area, velocity and discharge remain constant along the channel. Then the bed slope S_0, water surface slope S_w and energy-line slope S_f are all equal:

$$\mathbf{S_0 = S_w = S_f}$$

The flow depth is called **normal depth y_n**. It is reached only in long, prismatic, straight channels of constant roughness when gravity force equals boundary shear force.

Force balance on a control volume: $\gamma A L \sin\theta = \tau_0 P L \rightarrow \tau_0 = \gamma R S_0$.

2.2 Chezy equation

Equating shear stress $\tau_0 = k \rho V^2$ to $\gamma R S_0$ gives

$$\mathbf{V = C\sqrt{(R\,S_0)}}$$

C = Chezy's coefficient ($m^{1/2}/s$). Empirical: **Kutter** and **Bazin** formulae; $C = (1/n)R^{1/6}$ (Manning); $C = \sqrt{(8g/f)}$ (Darcy–Weisbach).

2.3 Manning's equation

The most widely used formula in SI units:

$$\mathbf{V = (1/n)\,R^{2/3}\,S_0^{1/2} \text{ and } Q = (1/n)\,A\,R^{2/3}\,S_0^{1/2}}$$

Surface	Manning n
Smooth concrete	0.012 – 0.014
Brick / rough concrete	0.015 – 0.017
Earth channel, clean	0.018 – 0.025
Earth with weeds	0.030 – 0.040
Natural stream, clean, straight	0.025 – 0.033
Mountain stream with boulders	0.040 – 0.070
Flood plain with trees	0.075 – 0.150

Conveyance $K = (1/n)\,A\,R^{2/3}$ so that $Q = K\sqrt{S_0}$.

2.3.1 Computation of normal depth

Given Q, n, S_0 and geometry, solve $A R^{2/3} = Q n/\sqrt{S_0}$ for y_n. This is non-linear, so use trial-and-error, charts, Newton–Raphson or software. For a **wide rectangular channel** $R \approx y$: $q = (1/n)\,y^{5/3}\,S_0^{1/2} \rightarrow y_n = (qn/\sqrt{S_0})^{3/5}$.

✎ **Exam Question:** A trapezoidal channel with b = 5 m, m = 1.5, n = 0.025, S_0 = 0.0004 carries 12 m^3/s. Find y_n. *Method: required $AR^{2/3} = 12 \times 0.025/0.02 = 15.0$; trial y = 1.5 m → A = 10.875, P = 10.41, R = 1.045, $AR^{2/3}$ = 11.2; y = 1.8 m → A = 13.86, P = 11.49, R = 1.206, $AR^{2/3}$ = 15.8; y ≈ 1.75 m.*

2.4 Most efficient (best hydraulic) section

For a given area A and slope, discharge is maximum when wetted perimeter P is minimum.

- **Rectangle:** b = 2y, R = y/2.

- **Trapezoid:** half of a regular hexagon, side slope 60° (m = 1/√3 = 0.577); R = y/2 for every efficient section.
- **Triangle:** side slope 45° (m = 1), R = y/(2√2).
- **Circle:** maximum velocity at y = 0.81 D; maximum discharge at y = 0.95 D.

★ **Key Point:** Best hydraulic section is not always the most economical: excavation, lining and construction practice must also be considered.

2.5 Compound channels

Divide the section into sub-areas (main channel + flood plains) by vertical lines (not counted in wetted perimeter), compute Q for each with its own n, and sum: $Q = \Sigma K_i \sqrt{S_0}$. Equivalent roughness (Horton/Einstein): $n_e = [\Sigma P_i n_i^{1.5}/P]^{2/3}$.

2.6 Design of non-erodible channels

Steps: (1) choose n for lining, (2) choose side slope m, (3) check maximum permissible velocity, (4) assume b/y ratio, (5) compute A from Q/V, (6) check Froude number, (7) add **freeboard** (typically 0.3–0.9 m, or 5–30% of depth).

3. SPECIFIC ENERGY

3.1 Definition

Specific energy E is the energy per unit weight of water measured with respect to the channel bed as datum:

$$\mathbf{E = y + \alpha V^2/2g = y + Q^2/(2\,g\,A^2)}$$

For a rectangular channel with discharge per unit width q = Q/b: $\mathbf{E = y + q^2/(2gy^2)}$.

3.2 Specific energy curve (for constant q)

E is a cubic in y: $y^3 - E\,y^2 + q^2/2g = 0$. For a given E there are generally three roots; one negative (meaningless) and two positive roots called **alternate depths** y_1 (supercritical, small) and y_2 (subcritical, large). At the minimum energy E_{min}, the two depths coincide at **critical depth y_c**.

- Upper limb of the curve: subcritical ($y > y_c$); lower limb: supercritical ($y < y_c$).
- Asymptotes: E = y (45° line) and y = 0 (the E axis).

3.3 Critical flow conditions

Set $dE/dy = 0$: $1 - Q^2T/(gA^3) = 0$, hence

$$\mathbf{Q^2/g = A^3/T \Leftrightarrow Fr = 1}$$

- **Rectangular:** $y_c = (q^2/g)^{1/3}$; $V_c = \sqrt{(g\,y_c)}$; $\mathbf{E_{min} = (3/2)\,y_c}$; $V_c^2/2g = y_c/2$.
- **Triangular:** $y_c = (2Q^2/(g\,m^2))^{1/5}$; $E_{min} = (5/4)y_c$.
- **Trapezoidal:** solve by trial: $Q^2/g = [(b + my_c)y_c]^3/(b + 2my_c)$.
- **Circular:** use charts or trial.

★ **Key Point:** Critical depth depends ONLY on discharge and channel geometry (not on slope or roughness). Normal depth depends on both. When $y_n > y_c$ the slope is **mild**; $y_n < y_c$: **steep**; $y_n = y_c$: **critical** slope S_c; $y_n \to \infty$: **adverse**; $S_0 = 0$: **horizontal**.

3.4 Alternate depths and transitions

Hump (raised bed Δz): $E_2 = E_1 - \Delta z$. For subcritical approach, y drops over the hump; for supercritical approach, y rises. If $\Delta z > E_1 - E_{min}$, **choking** occurs: upstream depth rises (subcritical) to supply the required energy and critical flow occurs on the hump crest.

Width contraction (rectangular): q increases from q_1 to $q_2 = Q/b_2$. Maximum contraction for no change upstream: $b_{2,min} = Q/[\sqrt{g}\,(2E_1/3)^{3/2}]$ (i.e. condition $E_1 = E_{min}$ at the throat).

3.5 Worked example

q = 3.0 m^2/s, y_1 = 2.0 m. Find alternate depth. V_1 = 1.5 m/s, $E_1 = 2.0 + 1.5^2/19.62 = 2.115$ m. $y_c = (9/9.81)^{1/3} = 0.972$ m. Solve $y + 0.4587/y^2 = 2.115$ for the small root by trial: y = 0.55 gives 2.066; y = 0.54 gives 2.113. So the alternate (supercritical) depth is $\mathbf{y_2 \approx 0.54\ m}$ (V = 5.56 m/s, Fr = 2.4).

4. SPECIFIC FORCE (MOMENTUM FUNCTION)

4.1 Concept

When energy losses are unknown (e.g. hydraulic jump) the **momentum equation** is used. The **specific force** (force per unit weight of water) at a section is

$$M = Q^2/(gA) + A\,\bar{y}$$

where $A\,\bar{y}$ is the first moment of the area about the free surface ($\bar{y}$ = depth of centroid below the water surface) and Q^2/gA is the momentum flux. For a rectangular channel: $M = q^2/(g\,y) + y^2/2$ per unit width.

4.2 Specific force curve

Plotting y against M for a constant Q gives a curve with a minimum M at **critical depth** ($dM/dy = 0$ gives the same condition $Q^2/g = A^3/T$). For a given M two depths exist called **conjugate (sequent) depths** y_1 and y_2.

Concept	Energy (specific energy)	Momentum (specific force)
Quantity conserved	Energy (frictionless)	Force/momentum
Depth pair	Alternate depths (E same)	Conjugate depths (M same)
Use	Smooth transitions, weirs	Hydraulic jump, flow through gates, forces on structures
Minimum at	Critical depth	Critical depth

4.3 Hydraulic jump (introductory)

A **hydraulic jump** is the abrupt transition from supercritical to subcritical flow with large energy dissipation; the momentum equation gives ($M_1 = M_2$):

$$y_2/y_1 = \tfrac{1}{2}\,[\,\sqrt{(1 + 8\,Fr_1^{\,2})} - 1\,] \text{ (rectangular channel)}$$

Energy loss: $h_L = (y_2 - y_1)^3/(4\,y_1\,y_2)$. Length of jump $L_j \approx 6\,y_2$ (5–7). Efficiency = E_2/E_1.

Fr_1	Type of jump	Energy dissipation
1.0 – 1.7	Undular	< 5%
1.7 – 2.5	Weak	5 – 15%
2.5 – 4.5	Oscillating	15 – 45%
4.5 – 9.0	Steady (best)	45 – 70%
> 9.0	Strong (choppy)	70 – 85%

4.4 Applications of specific force

- Stilling-basin design for dams and spillways.
- Jump below sluice gates; forces on gate: $F = \gamma(M_1 - M_2)$ shifted.
- Location of jump on sloping apron – use y_2 from jump equation vs tailwater depth: tailwater > y_2 drowns the jump; < y_2 sweeps it downstream.

✎ **Exam Question:** A rectangular channel carries 5 $m^3/s/m$ at $y_1 = 0.5$ m. Determine y_2, loss of head and power dissipated. *Ans: $V_1 = 10$ m/s, $Fr_1 = 4.52$, $y_2 = (0.5/2) \times (\sqrt{(1 + 8\times20.4)} - 1) = 2.95$ m, $h_L = (2.45)^3/(4\times0.5\times2.95) = 2.49$ m, $P = \gamma\,q\,h_L = 9.81 \times 5 \times 2.49 \approx 122$ kW per metre width.*

5. GRADUALLY VARIED FLOW (GVF) – THE DYNAMIC EQUATION

5.1 Assumptions

- Flow is steady; streamlines are nearly parallel (hydrostatic pressure).
- Bed slope is small ($\cos\theta \approx 1$, $\sin\theta \approx \tan\theta = S_0$).
- Channel is prismatic with constant α and roughness; head loss computed from uniform-flow formula using local V and R: $S_f = n^2V^2/R^{4/3}$.
- Discharge is constant along the reach (for SVF see Section 8).

5.2 Derivation

Total head $H = z + y + \alpha V^2/2g$. Differentiate with respect to x (distance along bed): $dH/dx = -S_f$ (energy slope is the negative of H gradient). Also $dz/dx = -S_0$. So

$-S_f = -S_0 + dy/dx + d/dx\,(\alpha Q^2/2gA^2)$.

Since $dA = T\,dy$, $d/dx(Q^2/2gA^2) = -(Q^2T/gA^3)\,dy/dx = -Fr^2\,dy/dx$. Therefore:

$$\mathbf{dy/dx = (S_0 - S_f) / (1 - Fr^2)}$$

This is the **dynamic equation of GVF**. Alternative forms with Manning's formula, using K = conveyance and K_n = normal conveyance, $Z = A\sqrt{D}$:

$$\mathbf{dy/dx = S_0\,[1 - (K_n/K)^2] / [1 - (Z_c/Z)^2]}$$

For a wide rectangular channel ($R \approx y$): $\mathbf{dy/dx = S_0\,[1 - (y_n/y)^{10/3}] / [1 - (y_c/y)^3]}$.

5.3 Interpretation of the equation

- Numerator > 0 when $y > y_n$ ($S_f < S_0$); < 0 when $y < y_n$.
- Denominator > 0 when $y > y_c$ ($Fr < 1$); < 0 when $y < y_c$.
- $dy/dx > 0$ means depth increases downstream. As $y \to y_n$, $dy/dx \to 0$ (flow becomes uniform – the profile is asymptotic to the normal depth line). As $y \to y_c$, $dy/dx \to \infty$ (the profile meets the critical depth line at a steep angle, and the GVF assumption breaks down).

6. CLASSIFICATION OF WATER-SURFACE PROFILES

6.1 Zones and slope types

Draw the normal depth line (NDL) and critical depth line (CDL). They divide the space above the bed into three zones:

- **Zone 1:** y above both lines.
- **Zone 2:** y between the lines.
- **Zone 3:** y below both lines.

Slopes: **M** (mild, $y_n > y_c$), **S** (steep, $y_n < y_c$), **C** (critical, $y_n = y_c$), **H** (horizontal, $S_0 = 0$), **A** (adverse, $S_0 < 0$). Profile name = slope letter + zone number (M1, M2, M3, S1, S2, S3, C1, C3, H2, H3, A2, A3).

6.2 Summary of the 12 profiles

Profile	Condition	dy/dx	Shape	Example
M1	$y > y_n > y_c$	+	Rising, asymptotic to horizontal at downstream end	Backwater upstream of dam
M2	$y_n > y > y_c$	−	Drawdown	Approach to a free overfall / change to steeper slope
M3	$y_n > y_c > y$	+	Rising, supercritical	Downstream of sluice gate on mild slope, ends in jump
S1	$y > y_c > y_n$	+	Rising to horizontal	Upstream of a dam/weir on steep slope; begins with jump
S2	$y_c > y > y_n$	−	Falling to y_n	Mild-to-steep slope break (downstream)
S3	$y_c > y_n > y$	+	Rising to y_n	Below a gate on steep slope
C1 / C3	$y > y_c = y_n$; $y < y_c$	+	Horizontal at y_c	Transition near critical slope
H2 / H3	$S_0 = 0$	− / +	Drawdown / rising	Flat apron / channel end
A2 / A3	$S_0 < 0$	− / +	Drawdown / rising	Adverse slope

★ **Key Point:** Zone 2 profiles are always drawdown curves (depth falls) and zone 1 and 3 are always rising (backwater) curves. Subcritical flow profiles (M1, M2, S1, H2, A2) are computed from the downstream control going upstream; supercritical flow profiles (M3, S2, S3, H3, A3) are computed from the upstream control going downstream.

6.3 Control sections

A **control** is a section where a known depth–discharge relation exists: critical depth at a free overfall or crest of a weir; sluice gate (vena contracta depth); reservoir (known level); normal depth far upstream/downstream.

✎ **Exam Question:** Sketch and explain the possible GVF profiles when a mild slope is followed by a steep slope; and when a steep slope is followed by a mild slope (jump). (8 marks)

7. FLOW PROFILE COMPUTATIONS

7.1 Direct step method (distance from depth)

For prismatic channels. Choose consecutive depths y_1, y_2 and compute the distance Δx. From energy balance between sections:

$$\Delta x = (E_2 - E_1) / (S_0 - S_f)$$

where $E = y + \alpha V^2/2g$ and $S_f = ½(S_{f1} + S_{f2})$ (or the Manning S_f at mean R and V). Steps: choose y_1 at the control; choose Δy; compute E, S_f at both sections; calculate Δx; Σ Δx gives position. Positive Δx in the downstream direction for supercritical (go downstream); negative for subcritical (go upstream). Advantage: no iterations. Limitation: only for prismatic channels; the distances are results, not inputs.

7.2 Standard step method (depth from distance)

Used for natural channels with irregular geometry. The stations x_i are fixed (cross-sections surveyed); depth/water level at the next section is found by trial:

$$z_1 + y_1 + \alpha_1 V_1^2/2g = z_2 + y_2 + \alpha_2 V_2^2/2g + h_f + h_e$$

with $h_f = S_f \Delta x$ and eddy/contraction loss $h_e = C_e|\alpha_2 V_2^2/2g - \alpha_1 V_1^2/2g|$ (C_e = 0.1 contraction, 0.3 expansion). Assume y_2, compute RHS, compare with LHS, correct and iterate until the balance error is < 0.003 m. This is exactly the algorithm in HEC-RAS.

7.3 Numerical integration of the dynamic equation

- **Euler / Predictor-corrector:** $y_{i+1} = y_i + \Delta x\, f(x_i, y_i)$.
- **Runge–Kutta 4th order:** more accurate, larger steps.
- **Bresse's method:** analytical solution for wide rectangular channels using the varied-flow function F(u, N) tables, with $u = y/y_n$:

$$x = (y_n/S_0)\,[\,u - F(u, N) + (y_c/y_n)^3\, F(u, 3)\,] + C$$

- **Chow's method:** uses varied-flow functions with hydraulic exponents N and M for general channels.

7.4 Worked example (direct step)

Rectangular channel b = 10 m, n = 0.015, S_0 = 0.0004, Q = 30 m^3/s. Trial gives $y_n \approx 1.85$ m; $y_c = (3^2/9.81)^{1/3} = 0.97$ m, so the slope is mild. A dam raises depth to 3.0 m (M1 profile). Find the distance to y = 2.5 m.

- At y = 3.0: A = 30, V = 1.00 m/s, R = 1.875, E = 3.051 m, $S_f = (0.015\times1.0)^2/1.875^{4/3} = 9.7\times10^{-5}$.
- At y = 2.5: A = 25, V = 1.20 m/s, R = 1.667, E = 2.573 m, $S_f = 1.64\times10^{-4}$.
- $S_f = 1.31\times10^{-4}$; Δx = (2.573 − 3.051)/(0.0004 − 0.000131) = −1774 m, i.e. the depth is 2.5 m about 1770 m **upstream** of the dam. Repeat for smaller steps; the profile approaches y_n asymptotically.

8. SPATIALLY VARIED FLOW (SVF)

8.1 Definition and types

SVF has non-uniform discharge along the channel, due to addition or withdrawal of water along its length.

- **Increasing discharge (lateral inflow):** roadside gutters, spillway side-channel, wash-water troughs in filters, drainage.
- **Decreasing discharge (lateral outflow):** side weirs, bottom racks (screens), irrigation canal with turnouts, perforated pipe.

8.2 Dynamic equation – increasing discharge

Apply the **momentum equation** (not energy, because mixing of the inflow causes unknown energy loss) to a CV of length dx:

$$\mathbf{dy/dx = [S_0 - S_f - (2Qq^*)/(gA^2)] / (1 - Q^2T/(gA^3)) \text{ where } q^* = dQ/dx}$$

(Assumes the lateral inflow enters at right-angle with no x-momentum.) Used for the side-channel spillway: depth increases downstream in the channel, critical section occurs at the end (control section, depends on friction).

8.3 Dynamic equation – decreasing discharge

Here the diverted water has the same specific energy as the main flow, so the **energy** approach is used (the specific energy is constant at the outflow location):

$$\mathbf{dy/dx = [S_0 - S_f - (\alpha Q/(gA^2))\, q^*] / (1 - \alpha Q^2T/(gA^3)) \text{ with } q^* = dQ/dx \text{ (negative)}}$$

Side weir: $q = (2/3)\, C_d \sqrt{(2g)}\, (y - w)^{3/2}$ per unit length (w = weir height). In subcritical approach the depth rises along the weir; in supercritical approach it falls.

8.4 Applications of SVF

- Side-channel spillway for dam with long crest (energy-dissipation and passing floods).
- Side weirs in sewers (combined sewer overflows) and irrigation canals to divert excess flow.
- Bottom racks / Coanda screens in hydropower intakes (sediment exclusion).
- Roof and street gutters; drainage channels in highways and airports; Wash-water troughs.
- Perforated pipe manifolds (decreasing discharge).

✎ **Exam Question:** Derive the dynamic equation for spatially varied flow with increasing discharge. State the assumptions. (10 marks)

9. RAPIDLY VARIED FLOW (RVF) – APPLICATIONS

9.1 Characteristics

In RVF the curvature of streamlines is large; pressure is non-hydrostatic; boundary friction is small compared with the effect of rapid curvature. The analysis uses energy and momentum principles with empirical coefficients.

9.2 Hydraulic jump – design applications

- **Stilling basins (USBR types I–IV):** Type I for $Fr_1 < 1.7$ (simple apron); Type II for Fr > 4.5 and V > 15 m/s (chute blocks, dentated sill); Type III for V < 15 m/s (baffle blocks and end sill); Type IV for Fr = 2.5–4.5 (oscillating jump).
- **Jump on a sloped floor:** longer jump, position unstable.
- **Jump as an energy dissipator and mixer:** chemical dosing in water treatment, aeration.
- **Submerged jump** (drowned): when tailwater depth > y_2; used for sluice-gate outlets.

9.3 Sluice gate (under-gate flow)

Discharge per unit width: $\mathbf{q = C_d\, a\, \sqrt{(2g\, y_1)}}$, a = gate opening, y_1 = upstream depth; $C_d \approx 0.6$ for free flow. Contraction coefficient $C_c \approx 0.61$: vena contracta depth $y_{vc} = C_c\, a$. Downstream flow is supercritical; a jump forms if the tailwater is high enough.

9.4 Flow over weirs, transitions and drops

- **Free overfall (brink):** critical depth occurs about 3–4 y_c upstream of the brink; **brink depth $y_b \approx 0.715\, y_c$** (rectangular). A way to measure Q.
- **Channel transitions:** contraction/expansion; the design uses specific-energy or Froude-number approach; for supercritical flow, oblique shock waves (hence curved or straight-walled design).
- **Drop structures and chute blocks.**
- **Culvert flow:** inlet control versus outlet control; headwater depth.

9.5 Choking and critical-flow control

A constriction can drive the flow to critical depth; discharge becomes dependent only on the upstream head, so the structure acts as a control (flume, weir).

10. INTRODUCTION TO HEC-RAS

10.1 What is HEC-RAS?

HEC-RAS (Hydrologic Engineering Center's River Analysis System) is free software of the U.S. Army Corps of Engineers for 1-D (and 2-D) hydraulic modelling of rivers and channels. Capabilities:

- Steady-flow water-surface profile computation (standard step method).
- Unsteady flow simulation (Saint-Venant equations, implicit finite-difference).
- Sediment transport/movable boundary computations.
- Water-quality analysis; 2-D flood inundation mapping (RAS Mapper).

10.2 Data required

Data type	Details
Geometry	River centreline, cross-sections (station–elevation), bank stations, reach lengths (left, channel, right)
Roughness	Manning n for left overbank, channel, right overbank
Structures	Bridges, culverts, weirs, inline structures (dams)
Flow data	Peak discharges for each profile; unsteady hydrographs
Boundary conditions	Known WS, critical depth, normal depth (slope), rating curve
Loss coefficients	Contraction 0.1, expansion 0.3

10.3 Workflow for steady-flow analysis

- 1. Start a new project; set units (SI).
- 2. Enter geometric data (reach schematic → cross-sections → structures).
- 3. Enter steady-flow data and boundary conditions (subcritical: downstream; supercritical: upstream; mixed: both).
- 4. Run steady-flow analysis (choose flow regime).
- 5. View results: profile plot, cross-section plots, tables (WS elevation, velocity, Fr, energy grade, top width).
- 6. Check errors/warnings (balance error, critical depth, conveyance ratio), calibrate n.
- 7. Floodplain mapping and report.

10.4 Theoretical basis in brief

- **Energy equation** solved by the standard step method between consecutive cross-sections.
- **Momentum equation** is used where flow is rapidly varied: hydraulic jumps, bridges, stream junctions.
- **Conveyance** K is computed by subdividing cross-sections at n-value changes. Average friction slope: $S_f = ((Q_1 + Q_2)/(K_1 + K_2))^2$ (average conveyance), or other options.
- **Unsteady flow:** Saint-Venant continuity $\partial A/\partial t + \partial Q/\partial x = 0$ and momentum $\partial Q/\partial t + \partial(\beta Q^2/A)/\partial x + gA(\partial y/\partial x + S_f - S_0) = 0$ solved by a four-point implicit (Preissmann) scheme.

10.5 Quick revision – formulae

Topic	Formula
Manning	$Q = (1/n)AR^{2/3}S_0^{1/2}$
Froude number	$Fr = V/\sqrt{(gA/T)}$
Specific energy	$E = y + Q^2/2gA^2$
Critical depth (rectangle)	$y_c = (q^2/g)^{1/3}$, $E_{min} = 1.5y_c$
Specific force	$M = Q^2/gA + A\bar{y}$
Conjugate depths	$y_2/y_1 = ½(\sqrt{(1 + 8Fr_1^2)} - 1)$
GVF equation	$dy/dx = (S_0 - S_f)/(1 - Fr^2)$
Direct step	$\Delta x = \Delta E/(S_0 - S_f)$